

Linear Algebra 1

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1 Introduction

1. If B is invertible and A is any matrix, $\text{rank}(A) = \text{rank}(AB) = \text{rank}(BA)$.
2. $\text{rank}(A) = \text{rank}(AA^T) = \text{rank}(A^T A)$. First equality follows from claim 3.
3. If $v_1, v_2, \dots, v_n$ are vectors of fixed finite dimension and $w \in \text{span}(v_1, \dots, v_n)$ such that $w^T \cdot v_i = 0$ for all $i \in [n]$, then $w = \vec{0}$.

Proof: $w = \sum c_i v_i$ for some c_i . Now $|w|^2 = w \cdot w = w \cdot \sum c_i v_i = \sum c_i w \cdot v_i = 0$, so $w = \vec{0}$.

4. To show $\text{rank}(A) = \text{rank}(A^T)$, ISTS $\text{rank}(A) = \dim(R(A)) = \dim(C(A))$ is well-defined. Claim 1 is then natural (should follow). The second equality of claim 2 follows. Both are the number of pivots after reducing. $\text{rank}(A) + \dim(N(A)) = m$ follows.

5. In an inner product space, pairwise nonzero orthogonal vectors are independent.

6. Let $v_1, v_2, \dots, v_n$ be nonzero pairwise orthogonal. Then $w \in \text{span}(v_i)$ iff $w = \sum \text{proj}_{v_i}(w)$.

7*. Let $v_1, v_2, \dots, v_n$ be independent. If there exists $w \neq \vec{0}$ such that $w = \sum \text{proj}_{v_i}(w)$, then the v_i are pairwise orthogonal.

8. If $W < V$ is a subspace of vector space V and $\dim W = \dim V < \infty$, then $W = V$.

9. If $v_1, v_2, \dots, v_n$ are n -dimensional nonzero orthogonal vectors, then $\sum \frac{v \cdot v^T}{v^T v} = I$.

10. If A has rank r , the projection matrix P_C onto $C(A)$ has $L(P_C) = r$.

11. If $B \sim A$, then $\text{char} B \equiv \text{char} A$. Proof:

$$\begin{aligned}\det(tI - B) &= \det(S^{-1}(tI - A)S) \\ &= \det(S^{-1}) \det(tI - A) \det(S) = \det(tI - A).\end{aligned}$$

It follows $\text{tr}(A) = \text{tr}(B)$ ($\det A = \det B$ is trivial).

12. Let square $n \times n$ matrix A have eigenvalues λ_i for $i \in [k]$. For each λ_i associate the space $N_i = N(\lambda_i I - A) \neq \{0\}$. Then the spaces N_i are independent.

Proof: Lemma.

$$\det \begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \dots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \lambda_2^2 & \dots & \lambda_2^{k-1} \\ \vdots & & \vdots & \ddots & \vdots \\ 1 & \lambda_k & \lambda_k^2 & \dots & \lambda_k^{k-1} \end{bmatrix} = \prod_{1 \leq i < j \leq n} (\lambda_j - \lambda_i).$$

We proceed by induction. The base case for $k = 2$ holds. Assume the result for k . Treat the determinant as a polynomial in λ_1 , a degree $k - 1$ polynomial in particular. The matrix is singular for $\lambda_1 = \lambda_2, \lambda_3, \dots, \lambda_k$ because of duplicate rows, so write

$$d(\lambda_1) = c(\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3) \cdots (\lambda_1 - \lambda_k)$$

for some constant c . Using cofactors and the inductive hypothesis,

$$c = (-1)^{k-1} \prod_{2 \leq i < j \leq n} (\lambda_j - \lambda_i).$$

Hence,

$$\begin{aligned}d(\lambda_1) &= (-1)^{k-1} \prod_{2 \leq i < j \leq n} (\lambda_j - \lambda_i) \prod_{j=2}^k (\lambda_1 - \lambda_j) \\ &= \prod_{1 \leq i < j \leq n} (\lambda_j - \lambda_i).\end{aligned}$$

This completes the inductive step.

As a corollary, if all of the λ 's are distinct (true since we count λ s of many multiplicity as one), then the determinant of that matrix (which we call P for powers) is nonzero.

Now suppose for some $v_i \in N_i$ there exists not all zero c_i for which $c_1 v_1 + c_2 v_2 + \dots + c_k v_k = \vec{0}$. Note that $(\lambda_i I - A)v_i = \vec{0}$, so $\lambda_i v_i = A v_i$. Multiplying the

equation by A , we get $c_1\lambda_1v_1 + c_2\lambda_2v_2 + \dots + c_k\lambda_kv_k = \vec{0}$. Repeat this process until we have $c_1\lambda_1^{k-1}v_1 + c_2\lambda_2^{k-1}v_2 + \dots + c_k\lambda_k^{k-1}v_k = \vec{0}$. This is to say

$$\begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{bmatrix} c_1 & & & \\ & c_2 & & \\ & & \ddots & \\ & & & c_k \end{bmatrix} \begin{bmatrix} 1 & \lambda_1 & \dots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \dots & \lambda_2^{k-1} \\ \vdots & & \ddots & \vdots \\ 1 & \lambda_k & \dots & \lambda_k^{k-1} \end{bmatrix} = 0_M,$$

$VCP = 0_M$. Since P is invertible, $VC = 0_M$, so $v_i c_i = \vec{0}$ for all $i \in [k]$. It is assumed that the v_i 's are nonzero, so all of the c_i 's are zero, and the v_i 's are independent.

13. Let A be symmetric. If v, v' are eigenvectors of $\lambda \neq \lambda'$, then v, v' are orthogonal.

$Av = \lambda v \rightarrow v^T A = \lambda v^T \rightarrow v^T A v' = \lambda v^T v' \rightarrow v^T \lambda' v' = \lambda v^T v' \rightarrow (\lambda - \lambda') v^T v' = 0 \rightarrow v^T v' = 0$, so the Eigenspaces are orthogonal. As a corollary if $\phi_A(x)$ has n distinct roots, then picking an eigenvector from each space gives an orthogonal basis of $\mathbb{R}^n$.

14. Suppose $\phi(x) = \det(xI - A)$. Let S_k denote the set of all size- k subsets of $[n]$. Then for all $i \in [n]$,

$$\phi^{(i)}(x) = i! \sum_{S \in S_{n-i}} (xI - A)(S \times S),$$

where $(xI - A)(S \times S)$ denotes the $(n - i) \times (n - i)$ submatrix of $xI - A$ by taking row and column indices in S . Induction step probably holds?

If A is orthonormal, then $\det A = \pm 1$. $\det(A)^2 = \det(A^T) \det(A) = \det(A^T A) = \det I = 1$. Elements of S_n are orthonormal.

Since there exists an embedding of $\mathbb{C}$ into $\mathbb{R}^{2 \times 2}$, any polynomial equation $\sum_{i=0}^n a_i A^i$, where $a_i \in \mathbb{R}$, $a_n \neq 0$, has at least n solutions in $\mathbb{R}^{2 \times 2}$ counting multiplicity.

$$\text{Spec}(A^n) = \text{Spec}(A)^n.$$

Let matrix M be Hermitian. Then $x^H M x \in \mathbb{R}$. Proof: This follows from

$$(x^H M x)^H = x^H M x^{HH} = x^H M x.$$

The quantity $x^H M x$ is scalar, so the H -operator is ordinary complex conjugation. The quantity was invariant, and therefore real.

Let $v_1, v_2, \dots, v_n$ be independent vectors in an inner-product space W . Suppose for all $w \in W$ the projection of w onto $C(V)$ is equal to the sum of the

projections of w onto v_i . Then the v_i are orthogonal.

Proof: Normalize the v_i . The projection matrix P onto $C(V)$ is $V(V^T V)^{-1} V^T$. The sum of the projection matrices is $\sum_{i=1}^n v_i v_i^T = V V^T$. Since the results of the projections is the same for all w , these matrices are the same. Then we have $V(V^T V)^{-1} V^T = V V^T \rightarrow V^T V(V^T V)^{-1} V^T = V^T V V^T \rightarrow V^T = V^T V V^T \rightarrow V^T V = (V^T V)^2 \rightarrow V^T V = I$. Hence, the v_i are orthogonal.

Let A be real symmetric. A is said to be positive definite if for all $x \neq \vec{0}$, $x^T A x > 0$. Claim: A is positive definite iff all of the eigenvalues of A are positive.

Suppose A is positive definite. Let $v \in \mathbb{R}^n$ be an eigenvector of eigenvalue λ . Then

$$v^T A v = v^T \lambda v = \lambda v^T v > 0,$$

so $\lambda > 0$. Now suppose all eigenvalues $\lambda > 0$. Take for convenience an orthonormal basis $\{v_i\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A . Let $(v \neq \vec{0}) \in \mathbb{R}^n$, so there exist $(c_i) \neq \vec{0}$ such that $v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$. Then

$$\begin{aligned} v^T A v &= \sum c_i v_i^T \sum c_i A v_i \\ &= \sum c_i v_i^T \sum c_i \lambda_i v_i \\ &= \sum \lambda_i c_i^2 v_i^T v_i \\ &= \sum \lambda_i c_i^2. \end{aligned}$$

$c_i \neq 0$ for some i , so $v^T A v \geq \lambda c_i^2 > 0$.

If A is positive definite iff $A = M^T M$ for some invertible M .

Proof: Let $A = Q \Lambda Q^T$. Since each $\lambda > 0$, let $D^2 = \Lambda$. Then $A = Q D D Q^T$. We check that $(Q D)^T = D^T Q^T = D Q^T$ and $D Q^T$ is invertible (since each $\sqrt{\lambda} > 0$, D^T is invertible).

If $A = M^T M$ for some invertible M , consider $v^T M^T M v = |M v|^2 > 0$ for $v \neq \vec{0}$, so A is positive definite.

The eigenvalues of a unitary matrix U lie on the unit circle.

Let $U^H U = I$, v be an eigenvector corresponding to λ . $U v = \lambda v \rightarrow v^H U^H = v^H \bar{\lambda} \rightarrow v^H U^H U v = v^H \bar{\lambda} U v \rightarrow v^H v = \lambda \bar{\lambda} v^H v$. Since $v^H v \neq 0$, $\lambda \bar{\lambda} = 1 \rightarrow |\lambda| = 1$.

If Q is a square $n \times n$ matrix such that $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$, then Q is orthogonal. Proof: We have $\|Qx\|^2 = \|x\|^2 \rightarrow x^T Q^T Q x = x^T x$. Let $A = Q^T Q$, which is symmetric and therefore diagonalizable. Suppose v is an

eigenvector of A . Then $v^T Av = v^T v \rightarrow v^T \lambda v = v^T v \rightarrow \lambda = 1$, since $v^T v \neq 0$. Hence, A must be similar to I , so $A = I$ and $Q^T Q = I$.